

SolarGen Forecast Model

SolarGen

Model version: recalibrated with measured history from 2026-05-04 through 2026-05-12

1 Purpose and Scope

SolarGen estimates hourly rooftop photovoltaic generation for a 10 kWp south-facing rooftop array in OHZ / Osterholz-Scharmbeck. The model is used by both the browser forecast and the local forecast-history database.

The model has two calibration layers:

- a clear-sky anchor day: 50.23 kWh measured on 2026-05-01;
- a weather-response recalibration period: measured forecast history from 2026-05-04 through 2026-05-12.

The hourly input data comes from Open-Meteo:

$$I_h, C_h, R_h, T_h$$

where I_h is global tilted irradiance in W/m², C_h is cloud cover in percent, R_h is precipitation in mm, and T_h is ambient temperature in °C. Daily inputs are precipitation sum, mean cloud cover, and sunshine duration.

2 Notation

Symbol	Meaning
h	local hour, 0 through 23
P	configured PV capacity in kWp, default 10
F	feed-in cap in kW, default 6
I_h	Open-Meteo global tilted irradiance, W/m ²
C_h	hourly cloud cover fraction, $C_h = \text{cloud_pct}_h/100$
R_h	hourly precipitation, mm
T_h	ambient air temperature, °C
S	calibration scale solved from the clear-sky anchor day

All hourly kW averages are treated as kWh over that hour.

3 Weather-Adjusted Irradiance

3.1 Low-Irradiance Cloud Uplift

Very dark overcast days produced more rooftop output than raw tilted irradiance implied. The model therefore boosts low-irradiance cloudy hours:

$$L_h = \left[\text{clamp} \left(1 - \frac{I_h}{577}, 0, 1 \right) \right]^{1.855}$$

$$M_h = \min(1.748, 1 + 2.2C_h L_h)$$

Clear hours keep $M_h = 1$. The boost is strongest when cloud cover is high and irradiance is low.

3.2 Daily Weather Damping

The May 12 actual showed the opposite failure mode: bright hourly irradiance under wet, low-sun, high-cloud daily conditions can over-forecast. The model computes:

$$D_{\text{rain}} = 0.195 \cdot \text{smoothstep}(0, 3.116, R_{\text{day}})$$

$$D_{\text{sun}} = 0.747 \cdot \text{smoothstep}(10.267, 3.755, H_{\text{sun}})$$

$$D_{\text{cloud}} = 0.458 \cdot \text{smoothstep}(61.016, 93.522, C_{\text{day}})$$

$$D_{\text{day}} = \text{clamp}(D_{\text{rain}} + D_{\text{sun}} + D_{\text{cloud}}, 0, 1.2)$$

where R_{day} is daily precipitation in mm, H_{sun} is sunshine duration in hours, and C_{day} is mean daily cloud cover in percent.

3.3 Bright High-Cloud Damping

The bright-hour weight is:

$$B_h = \text{smoothstep}(265, 818, I_h)$$

The bright-cloud damping factor is:

$$Q_h = \text{clamp}(1 - 0.763C_h B_h D_{\text{day}}, 0.25, 1.1)$$

3.4 Hourly Rain Damping

$$U_h = \frac{1}{1 + 0.955R_h}$$

3.5 Adjusted Irradiance

The weather-adjusted irradiance is:

$$I'_h = I_h \cdot M_h \cdot Q_h \cdot U_h$$

4 Temperature Derating

Panel temperature is approximated from adjusted irradiance and ambient temperature:

$$T_{\text{panel},h} = T_h + \frac{I'_h}{800} \cdot 20$$

The derating factor is:

$$A_h = \text{clamp}(1 - 0.0035(T_{\text{panel},h} - 25), 0.82, 1.06)$$

5 Theoretical PV Output

Before applying the site-specific rooftop profile:

$$G_h^{\text{theory}} = \frac{I'_h}{1000} \cdot P \cdot S \cdot A_h$$

The calibration scale S is solved numerically so that the clear-sky irradiance curve for 2026-05-01 produces exactly 50.23 kWh after the rooftop profile is applied.

6 Rooftop Profile

The rooftop profile maps theoretical generation to the observed installation behavior. It models morning limitation, the main production window, evening drop-off, and diffuse-light behavior under cloud.

6.1 Clear-Sky Branch

The low-output branch is:

$$G_h^{\text{low}} = \min \left(0.286 G_h^{\text{theory}}, 1.44 \right)$$

The clear/direct branch is:

$$G_h^{\text{sun}} = \begin{cases} G_h^{\text{low}}, & h < 10.67 \\ G_h^{\text{theory}}, & 10.67 \leq h < 16.72 \\ (1 - p_h) G_h^{\text{theory}} + p_h G_h^{\text{low}}, & 16.72 \leq h < 17.61 \\ G_h^{\text{low}}, & h \geq 17.61 \end{cases}$$

where

$$p_h = \frac{h - (17.61 - 0.89)}{0.89}$$

6.2 Cloud/Diffuse Branch

Clouds increase the diffuse-light share:

$$W_h = \text{smoothstep}(57.94, 93.47, 100C_h)$$

The diffuse branch uses the hour center $x_h = h + 0.5$:

$$G_h^{\text{diff}} = G_h^{\text{theory}} \cdot \text{smoothstep}(4.5, 7.9, x_h) \cdot [1 - \text{smoothstep}(13.18, 19.9, x_h)] \cdot 1.2$$

6.3 Rooftop Generation Before Curtailment

$$G_h = (1 - W_h) G_h^{\text{sun}} + W_h G_h^{\text{diff}}$$

For hours 5 and 6 only, a small dawn diffuse floor is applied:

$$G_h = \max(G_h, 0.2C_h \cdot P/10)$$

7 Curtailment and Delivered PV

The feed-in cap is applied to hourly rooftop generation:

$$G_h^{\text{curt}} = \max(0, G_h - F)$$

$$G_h^{\text{deliv}} = G_h - G_h^{\text{curt}}$$

Daily rooftop generation is:

$$G_{\text{day}} = \sum_{h=0}^{23} G_h$$

8 Household Load and Battery Accounting

The default household load is:

$$L_h^{\text{home}} = L_{\text{base}} + \mathbf{1}_{8 \leq h < 18} L_{\text{day}} + \mathbf{1}_{18 \leq h < 23} L_{\text{eve}} + \mathbf{1}_{6 \leq h < 8} (0.45 L_{\text{day}})$$

PV first covers same-hour load. Remaining load is covered from the battery if available. PV surplus charges the battery with 94 percent charge efficiency. Remaining surplus is exported.

9 Calibration Procedure

Calibration is split into a structural clear-sky fit and a weather-response fit. This keeps the model identifiable: the clear-sky day fixes the site scale and rooftop geometry, while the historical cloudy and rainy days tune how meteorology changes production.

9.1 Clear-Sky Scale

Let A be the anchor day, 2026-05-01, with measured production

$$Y_A = 50.23 \text{ kWh.}$$

For any trial scale S , the model produces an anchor-day total

$$\hat{Y}_A(S) = \sum_{h=0}^{23} G_h(S).$$

The scale is solved numerically as the one-dimensional root

$$S^* = \arg \min_S \left(\hat{Y}_A(S) - Y_A \right)^2, \quad \hat{Y}_A(S^*) = Y_A.$$

This step fixes the absolute output level for the 10 kWp array after temperature derating and the rooftop profile.

9.2 Weather-Response Fit

The weather-response parameters are fitted against measured daily totals from 2026-05-04 through 2026-05-12. For each measured day d , let Y_d be actual generation and $\hat{Y}_d(\theta)$ be the modeled total under parameter vector θ :

$$\theta = (\theta_{\text{cloud}}, \theta_{\text{rain}}, \theta_{\text{sun}}, \theta_{\text{profile}}).$$

The fitting objective minimizes daily error while keeping the anchor scale fixed:

$$\theta^* = \arg \min_{\theta} \sum_{d \in \mathcal{D}} w_d \left(\hat{Y}_d(\theta) - Y_d \right)^2, \quad \mathcal{D} = \{2026-05-04, \dots, 2026-05-12\}.$$

Days with incomplete actuals are excluded from $\mathcal{D}$. The calibrated parameters primarily control four effects:

- low-irradiance cloud uplift, M_h , which raises modeled output on dark diffuse days;
- daily weather damping, D_{day} , which suppresses over-optimistic bright-hour forecasts on wet, low-sun, high-cloud days;
- hourly rain damping, U_h , which applies immediate precipitation loss;
- rooftop timing parameters, which encode the observed morning limitation and evening drop-off.

9.3 Model Selection Criterion

The selected parameter set is the simplest candidate that reproduces the historical totals without removing the physical structure of the model. The primary score is daily mean absolute error:

$$\text{MAE} = \frac{1}{|\mathcal{D}|} \sum_{d \in \mathcal{D}} \left| \hat{Y}_d - Y_d \right|.$$

Relative accuracy is tracked with daily mean absolute percentage error:

$$\text{MAPE} = \frac{100}{|\mathcal{D}|} \sum_{d \in \mathcal{D}} \left| \frac{\hat{Y}_d - Y_d}{Y_d} \right|.$$

The calibration period must be updated in this document and in the app header whenever the weather-response fit is regenerated.

10 Calibration Quality

Replaying the measured days from 2026-05-04 through 2026-05-12 gives:

Metric	Value
Days compared	7
Actual total	235.33 kWh
Model total	238.13 kWh
Daily mean absolute error	0.44 kWh
Daily RMSE	0.55 kWh
Daily MAPE	1.35 percent
Worst daily absolute percentage error	2.43 percent

These are in-sample statistics because the measured days were used for recalibration. Future actuals should be used as out-of-sample checks.

11 Implementation References

The corresponding implementation lives in:

- `src/config.js`: constants and calibration parameters;
- `src/model.js`: weather adjustment, rooftop profile, storage, export, and totals;
- `tests/model.test.mjs`: calibration replay tests and model invariants.